## Practice Problems 1

Not to be turned in, but may be helpful before the exam

1. On Homework 4, you worked with the random variables $Y_1, \ldots, Y_n$, where

$$Y_i = \beta x_i + e_i, \quad i = 1, \ldots, n$$

with $x_1, \ldots, x_n$ fixed constants and $e_1, \ldots, e_n$ iid $N(0, \sigma^2)$ random variables, with $\sigma^2$ unknown. On HW4, you found the MLE of $\beta$, along with its distribution. Now consider the estimator $\tilde{\beta} = \sum_i Y_i / \sum_i x_i$. Which is a better estimator, $\hat{\beta}_{\text{MLE}}$ or $\tilde{\beta}$? Explain your answer, citing relevant statistical evidence.

2. Also on Homework 4, you let $Y_1, \ldots, Y_n$ be iid random variables from the population with pdf

$$f(y; \beta) = \begin{cases} 3\beta^3 y^{-4} & y \geq \beta \\ 0 & \text{elsewhere} \end{cases}$$

and found the MLE of $\beta$.

   a. Find the bias of $\hat{\beta}_{\text{MLE}}$.
   b. Find MSE($\hat{\beta}_{\text{MLE}}$).

3. The *relative efficiency* is a way to compare the variances of two unbiased estimators. Suppose we have two estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, which are both unbiased for estimating $\theta$. The efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is:

$$\text{Efficiency}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Variance}(\hat{\theta}_2)}{\text{Variance}(\hat{\theta}_1)}$$

   a. Why do you think both estimators must be unbiased?
   b. Suppose that $X_1, \ldots, X_n$ are a random sample from the distribution with pdf $f(x|\theta) = e^{-(x-\theta)}$ for $x > \theta$.
      i. Find an estimator $\hat{\theta}_1$ for $\theta$ using the method of moments. Is $\hat{\theta}_1$ unbiased?
      ii. Find an estimator $\hat{\theta}_2$ for $\theta$ using the method of maximum likelihood. Is $\hat{\theta}_2$ unbiased?
      iii. Adjust $\hat{\theta}_1$ and $\hat{\theta}_2$ so that they are unbiased. Find the efficiency of the adjusted $\hat{\theta}_1$ relative to the adjusted $\hat{\theta}_2$. What do you conclude?

4. Consider a random sample $X_1, \ldots, X_n$ from a uniform$(\theta_1, \theta_2)$ distribution. We are interested in estimating the mean of the uniform$(\theta_1, \theta_2)$ distribution, $\mu = \frac{\theta_1 + \theta_2}{2}$.

   a. Two different estimators have been proposed:

$$\hat{\mu}_1 = \bar{X} \qquad \text{and} \qquad \hat{\mu}_2 = \frac{X_{(1)} + X_{(n)}}{2}.$$

   Show that both $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators for $\mu$.
   b. WITHOUT calculating the variances of $\hat{\mu}_1$ and $\hat{\mu}_2$, which estimator do you think will be better? **Thoroughly** explain and justify your answer, using appropriate statistical evidence.

5. Let $X_1, \ldots, X_n$ be a random sample from the *Rayleigh* distribution (with parameter $\theta$). The pdf of the Rayleigh is:

$$f(x; \theta) = \left(\frac{2x}{\theta}\right) e^{-x^2/\theta}$$

for $x > \theta$. Show that $\sum_i X_i^2$ is sufficient for $\theta$.

6. Let $X_1, \ldots, X_n$ be a random sample from the *Pareto* distribution (with parameters $\alpha > 0$ and $\beta > 0$). The pdf of the Pareto is:

$$f(x; \alpha, \beta) = \alpha \beta^{\alpha} x^{-(\alpha+1)}$$

for $x \geq \beta$. Assuming that $\beta$ is known, show that $\prod_i X_i$ is sufficient for $\alpha$.

7. Let $X_1, \ldots, Xn$ be a random sample from the Uniform$(\theta_1, \theta_2)$ distribution. Show that $X_{(1)}$ and $X_{(n)}$ are jointly sufficient for $\theta_1$ and $\theta_2$.

8. Let $X$ be a single observation from the binomial$(n, p)$ distribution.

   a. Find the MLE of $p$, $\hat{p}_{\text{MLE}}$. Be sure to show/explain all work.
   b. Show that $\hat{p}_{\text{MLE}}$ is an unbiased estimator of $p$.
   c. To estimate the variance of $X$, we generally use $n\hat{p}_{\text{MLE}}(1 - \hat{p}_{\text{MLE}})$. Show that this typical estimator is a biased estimator of Var$(X)$.